ADVANTEST

ADVANTEST CORPORATION

ASX/U-51
SYSTEST TESTUTL ATLworks
UPDATE NOTICE
REV 6.02

MANUAL NUMBER 8262527-57

Applicable Products

*PASX51-93*****
*PASX51-92*****
*PASX51-91*****
*PASX51-85*****
*PASX51-81*****
*PASX51-75*****
*PASX51-71*****
*PASX51-70*****
*PASX51-36*****

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PREFACE

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RECORD OF REVISIONS

Manual Rev	Date	Remarks (Software REV)	Manual Rev	Date	Remarks (Software REV)
49a	Sep 25/01	(REV 6.00)			
50	Oct 18/01	(REV 6.00-1)			
51	Dec 26/01	(REV 6.00-2)			
52	Jan 9/02	(REV 6.00A)			
53	Feb 25/02	(REV 6.00A1)			
54	Jul 15/02	(REV 6.00B)			
55	Nov 1/02	(REV 6.00C)			
55a	Mar 3/03	(REV 6.00C1)			
56	Mar 7/03	(REV 6.01)			
57	May 15/03	(REV 6.02)			

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1. SUMMARY OF REVISION

(1) Revision number

- ASX/U-51 REV 6.02

(2) Applicable systems

- T5593
- T5592
- T5591/T5591R
- T5586/T5585
- T5581/T5581H
- T5581P/T5571P
- T5376/T5375
- T5371
- T5336/T5311

(3) Related manual revised

• ATL-51 TEST PLAN PROGRAM REFERENCE MANUAL	8431498-04
• ATL-51 SOCKET PROGRAM REFERENCE MANUAL	8256293-19
• ATL-51 SCRAMBLE PROGRAM REFERENCE MANUAL	8256294-16
• ATL-51 PATTERN PROGRAM REFERENCE MANUAL (T5500)	8256292-26
• ATL-51 PATTERN PROGRAM REFERENCE MANUAL (T5376/T5375/T5371/T5336/T5311)	8289058-15
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• ASX/U-51 HYBRID CONTROLLER REFERENCE MANUAL	8409600-06
• ASX/U-51 Communication Control Library Manual	8431496-01
• ERROR CODE TABLE REFERENCE MANUAL	8141091-73
• ADVANsite System Introduction	8227727-06
• ADVANsite TEST SYSTEM SOFTWARE INSTALLATION/SYSTEM START USER'S MANUAL VOL1 BASIC OS INSTALLATION	8227728-20
• ADVANsite TEST SYSTEM SOFTWARE INSTALLATION/SYSTEM START USER'S MANUAL VOL2 TESTER ENVIRONMENT SOFTWARE INSTALLATION	8227729-20

- Online Manual Reference Manual 8431523-05
- Online Manual Reference Manual (HTML) 8700036-01

(4) Reason of revision

- Addition of functions
- Improvement of functions

(5) Products revised

Products revised in this revision are shown below:

Product code	Product name		
PASX51-93****	SYSTEST51SU	TESTUTL51SU	ATW51SU
PASX51-92****			
PASX51-91****			
PASX51-85****			
PASX51-81****			
PASX51-75****			
PASX51-71****			
PASX51-70****			
PASX51-36****			

(6) List of revisions

The revision list shows the revision correspondence of the basic software the SYSTEST51SU, TESTUTL51SU and ATW51SU of ASX/U-51.

<Explanation of the revisions' list>

(a) "*" shows the product revised in the revision.

(b) "–" shows the product that does not exist to the whole revision.

ASX/U-51				
ASX/U-51 Overall Revision	Revision for each product			SunOS Revision
	SYSTEST51SU	TESTUTL51SU	ATW51SU	
5.04A	5.04A *	5.04A *	–	Solaris 2.5.1 Solaris 7
5.04A1	5.04A1 *	5.04A1 *	–	Solaris 2.5.1 Solaris 7
5.04A2	5.04A2 *	5.04A1	–	Solaris 2.5.1 Solaris 7
5.04A3	5.04A3 *	5.04A1	–	Solaris 2.5.1 Solaris 7
5.04A4	5.04A4 *	5.04A2 *	–	Solaris 2.5.1 Solaris 7
5.05(Note1)	5.05 *	5.05 *	5.05 *	Solaris 8
6.00	6.00 *	6.00 *	6.00 *	Solaris 2.5.1 Solaris 7 Solaris 8
6.00-1	6.00-1 *	6.00-1 *	6.00-1 *	Solaris 2.5.1 Solaris 7 Solaris 8
6.00-2	6.00-2 *	6.00-1	6.00-1	Solaris 2.5.1 Solaris 7 Solaris 8
6.00A	6.00A *	6.00A *	6.00A *	Solaris 2.5.1 Solaris 7 Solaris 8
6.00A1	6.00A1 *	6.00A1 *	6.00A	Solaris 2.5.1 Solaris 7 Solaris 8

ASX/U-51				
ASX/U-51 Overall Revision	Revision for each product			SunOS Revision
	SYSTEST51SU	TESTUTL51SU	ATW51SU	
6.00B	6.00B *	6.00B *	6.00B *	Solaris 2.5.1 Solaris 7 Solaris 8
6.00C	6.00C *	6.00C *	6.00C *	Solaris 2.5.1 Solaris 7 Solaris 8
6.00C1	6.00C1 *	6.00C1 *	6.00C	Solaris 2.5.1 Solaris 7 Solaris 8
6.01(Note2)	6.01 *	6.01 *	6.01 *	Solaris 8
6.02	6.02 *	6.02 *	6.02 *	Solaris 2.5.1 Solaris 7 Solaris 8

Note1: ***REV 5.05 is used for the T5375.***

Note2: ***REV 6.01 is used for the T5593.***

2. ENHANCED FUNCTIONS

2.1 Tester Driver Controller

2.1.1 32-Character Compatibility with ATL-51 Names

A maximum of 32 characters can now be specified for the names shown below.

In addition, usable character types have been added.

- ATL program

Name	Number of characters	Character type
File name	32	Alphanumeric characters, $, and . (point)
Customer name		
Program name		
Variable name	32	Alphanumeric characters beginning with an alphabetic character, and _ (under bar)
Label name		
EQUATE statement The character string which can be replaced	32	Alphanumeric characters beginning with an alphabetic character, - (hyphen), and _ (under bar)
EQUATE statement Argument character string	32	Alphanumeric characters beginning with an alphabetic character, and _ (under bar)
DMACRO statement Macro definition name	32	Alphanumeric characters beginning with an alphabetic character, - (hyphen), and _ (under bar)

- Socket program

Name	Number of characters	Character type
Socket program name	32	Alphanumeric characters, $, and . (point)
DUT pin name	32	Alphanumeric characters beginning with an alphabetic character, and _ (under bar)
Conversion symbol name		

- Scramble program

Name	Number of characters	Character type
Scramble program name	32	Alphanumeric characters, $, and . (point)

- Pattern program

Name	Number of characters	Character type
Pattern program name	32	Alphanumeric characters, $, and . (point)

The following programs have been changed due to the enhancement of this function:

- Tester driver controller
- Tester utility program
  Pin monitor (/PM)
  Data set (/Unit name) (*2)
  Debugger (/DEBUG)
  2-dimension shmoo plot (SHM2)
  3-dimension shmoo plot (SHM3)
  Pattern tracer (/MTRACE)
  Data scan (/DS)
  Program call (/MLOAD)
  User area display (/MLIST)
  Specifying the BUFF program (/BUFF)
  Executable program (/PRO)
  Histogram creation program (HIDAR)
  Wafer map
  Data log (/DTLOG)
  Partial test (/PART)
  Error code utility (/ERR)
  Clock setting program (/TGSET)
  Extension pin monitor (/EPM)
  MARGIN program (/MARGIN)
  LOG ON utility (/LOG ON, /LOG OFF)
  Program which controls the handler by using the GPIB interface (/STHAND)
  FORK
  Prober selection and initial setting program (/PROBSEL)
  CALCHECK
  GETHIDAR utility
  Pin monitor for each MEAS statement (/PMSET)
  Operation panel
  Fail memory data transfer (/FMXFER)
  Category and sort data output program (lcommand)
  ENDPRO
  Pulse count measurement utility (PULSECNT)
  ATLworks (*1) (*3)
- Controller commands
  O command
  S command
  I command
  J command
  V command
  .S command
  .T command
  L command
  A command
  T command
  P command
  CTRL/B command
  CTRL/T command
  U command

Relevant systems:	T5593, T5592, T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5376, T5375, T5371, T5336, and T5311
Compatible compilers:	XATL/U-51(PXTL51-00S***) REV 6.02 and later XATL/U-51(PXTL51-36S***) REV 3.01and later

Compatible Optional Software, System Diagnostic Programs, and Calibration Programs:

MRA4	REV 3.01A and later
Workbench-M2	REV 5.04 and later
Wave Tracer2	REV 5.04 and later
MPATsim	REV 5.03 and later
WFBMAP3	REV 1.03A and later
TCC/U-51	REV 2.00 and later
Tester Diagnosis (T5593)	REV 1.01 and later
Tester Diagnosis (T5592)	REV 1.06 and later
Tester Diagnosis (T5591)	REV 2.03 and later
Tester Diagnosis (T5585)	REV 2.04 and later
Tester Diagnosis (T5581/T5581H)	REV 3.03 and later
Tester Diagnosis (T5581P/T5571P)	REV 3.03 and later
Tester Diagnosis (T5375/T5376)	REV 1.03 and later
Tester Diagnosis (T5371)	REV 1.03 and later
Tester Diagnosis (T5336/T5311)	REV 1.06 and later
Precision Calibration (T5581/T5581H)	REV 1.04 and later
Precision Calibration (T5581P/T5571P)	REV 1.04 and later
Precision Calibration 2 (T5585/T5586)	REV 1.04 and later
Offset Calibration (T5585/T5586)	REV 1.02 and later

(*1) ***If ATLworks that is compatible with ASX/U-51 REV 6.02 is installed in a test system in which optional software (Workbench-M2, Wave Tracer2, MPATsim, and MRA4) that is incompatible with ASX/U-51 REV 6.02 is installed, the optional software buttons that should be displayed in the Tool Box of ATLworks disappear and the optional software cannot be started from ATL works.***

(*2) ***Data set (/unit name) cannot be executed with a test system in which XATL/U-51 that is incompatible with ASX/U-51 REV 6.02 is installed.***

(*3) ***If XATL/U-51 that is incompatible with ASX/U-51 REV 6.02 is installed in the test system, no settings can be changed in the Test Condition Monitor.***

Because of the enhancement of this function, the ASX/U-51 REV 6.02 and earlier (non- 32-character compatible version) and REV 6.02 and later (32-character compatible version) run differently under the following conditions:

(1) If a program, whose name exceeds the maximum number of characters, is compiled by using a compiler that is incompatible with 32-character names, the number of characters that exceed the maximum number is ignored. However, if the same program is compiled by using a compiler that is compatible with 32-character names, all characters including the excess number of characters are recognized and output to the object. Therefore, a non-32-character compatible version and a 32-character compatible version run differently as shown in the examples below.

Example 1:

If file, customer, and program names exceed 8 characters respectively, a non-32-character compatible version ignores all the remaining characters from the 9th character. In contrast, a 32-character compatible version recognizes up to 32 characters.

A program, which can be executed in a non-32-character compatible version, generates a runtime error if it is executed in a 32-character compatible version.

	Executes an object, which is incompatible with 32 characters.	Executes an object, which is compatible with 32 characters.
PRO SAMPLE1		
:		
CALL SUB456789	Calls SUB45678.	An error occurs because SUB456789 does not exist.
OPEN! CH R DAT456789	Opens DAT45678.	An error occurs because DAT456789 does not exist.
SEND SCRDATA SCR456789	Transfers SCR45678.	An error occurs because SCR456789 does not exist.
SEND MPAT MPA456789	Transfers MPA45678.	An error occurs because MPA456789 does not exist.
SEND DBM MPA456789	Transfers MPA45678	An error occurs because MPA456789 does not exist.
SEND FM MPA456789	Transfers MPA45678	An error occurs because MPA456789 does not exist.
MEAS MPAT MPA456789	Transfers MPA45678	An error occurs because MPA456789 does not exist.
:		
STOP		
END		

* This is an example when sub45678.sub, dat45678.dat, scr45678.scr, and mpa45678.mpa exist.

Example 2:

If a variable name exceeds 6 characters, a non-32-character compatible version ignores all the remaining characters from the 7th character. In contrast, a 32-character compatible version recognizes up to 32 characters.

A variable, whose type is declared in a non-32-character compatible version, may be treated as an unspecified type variable a 32-character compatible version.

PRO SAMPLE2	Executes an object, which is in-compatible with 32 character.	Executes an object, which is compatible with 32 characters.
:		
INTE[2] VAR456789	Declares a VAR456 type.	Declares a VAR456789 type.
VAR456=1	VAR456 is an INTE type.	VAR456 is an unspecified type.
:		
STOP		
END		

Example 3:

If a label name exceeds 6 characters, a non-32-character compatible version ignores all the remaining characters from the 7th character. In contrast, a 32-character compatible version recognizes up to 32 characters.

A program, which can be compiled in a non-32-character compatible version, generates a compilation error if it is compiled in a 32-character compatible version.

PRO SAMPLE3	Compiled by using a non-32-character compatible version compiler.	Compiled by using a 32-character compatible version compiler.
:		
L23456:		
:		
GOTO L23456789	The GOTO statement, where the program branches out to L23456.	Compilation error
:		
STOP		
END		

Example 4:

If the number of characters of the argument character string in the EQUATE statement exceeds 8 characters, a non-32-character compatible version ignores all the remaining characters from the 9th character. In contrast, a 32-character compatible version recognizes up to 32 characters

A program, which can be compiled in a non-32-character compatible version, generates a compilation error if it is compiled in a 32-character compatible version.

```
PRO SAMPLE4
:
EQUATE W(A2345678) = @
"WRITE ?" A23456789 "?"
:
STOP
END
```

Non-32-character compatible version compiler.	32-character compatible version compiler.
A2345678 is the argument character string of A23456789.	Compilation error

(2) When using the controller command or the tester utility program, where a file name, customer name, and program name can be specified, a non-32-character compatible program automatically terminates when 8 characters are entered, and a 32-character compatible program terminates when 32 characters are entered.

Therefore, when using the KCALL statement to enter characters, if the file, customer, and program names are each 8 characters long, no carriage return is needed in a non-32-character compatible version. However, a carriage return is needed in a 32-character compatible version.

Example 1:

When entering an 8-character program name to MODE 1(PRO NAME) of the O command by using the KCALL statement, a non-32-character compatible program automatically terminates when 8 characters are entered. However, a 32-character compatible program does not automatically terminate when 8 characters are entered. Therefore, a carriage return is needed to terminate the program.

```
PRO SAMPLE5
:
KCALL ?O11PRO45678....?
:
:
STOP
END
```

Non-32-character compatible version	32-character compatible version
The program automatically terminates when MODE 1 is entered after PRO45678.	The program does not terminate unless a carriage return is entered after PRO45678.

The target controller commands and tester utility programs are as follows:

Fail memory data transfer (/FMXFER)

Partial test (/PART)

Program which controls the handler by using the GPIB interface (/STHAND)

O command

U command

2.1.2 Compatibility with the 2-Gigabits PM Board

The 2-Gigabits PM board can now be used.

For more information, refer to “ATL-51 TEST PLAN PROGRAM REFERENCE MANUAL.”

The following programs have been changed due to the enhancement of this function:

- Tester driver controller
- System information read utility (RDSYSTEM)

Relevant systems: T5586, T5585, T5581, T5581H, T5581P, and T5571P

2.2 Hybrid Controller

A C program that includes the ATL can now be used.

As a result, an ATL statement can now be written in a C program.

For more information, refer to “ASX/U-51 HYBRID CONTROLLER USER’S MANUAL.”

The following programs have been changed due to the enhancement of this function:

- Tester driver controller
- C compiler (atcc)

Relevant systems: T5593, T5592, T5586, T5585, T5376, T5375, and T5371

2.3 Communication Control Library

The communication control library has been added.

By using this library, the tester, handler, and prober can now be controlled from the EWS.

For more information, refer to “ASX/U-51 Communication Control Library Manual.”

Relevant systems: T5593, T5592, T5586, T5585, T5376, and T5375

2.4 Tester Utility Program

2.4.1 Auto Scan Mode Utility (AUTOSCAN)

The auto scan mode utility has been added to read the AUTOSCAN COUNTER.

For more information, refer to “T5500 SERIES TESTER UTILITY PROGRAM REFERENCE MANUAL.”

Relevant systems: T5593

2.4.2 RON/ROF Monitor Utility

The RON/ROF monitor utility has been added to monitor the VS reference ON and OFF state.

For more information, refer to “T5500 SERIES TESTER UTILITY PROGRAM REFERENCE MANUAL.”

Relevant systems: T5593, T5592, T5586, T5585, T5376, T5375, and T5371

2.5 ATLworks

2.5.1 DBMtool

(1) The DBMtool has been added to debug the DBM pattern program.

For more information, refer to “ATLworks TOOL URER’S MANUAL.”

Relevant systems: T5592

(2) The function, which is used to select a pattern program that is transferred to the DBM, has been added.

For more information, refer to “ATLworks TOOL URER’S MANUAL.”

Relevant systems: T5593

2.5.2 dbm2asc

The dbm2asc command has been added to convert the DBM pattern data in the MPAT object file to the ASCII data.

For more information, refer to “ATLworks TOOL URER’S MANUAL.”

Relevant systems: T5592

2.5.3 ASXDebugger

(1) A C program that includes the ATL can now be used.

For more information, refer to “ATLworks TOOL URER’S MANUAL.”

Relevant systems: T5593, T5592, T5586, T5585, T5376, T5375, and T5371

(2) The RON/ROF monitor can now be used.

For more information, refer to “ATLworks TOOL URER’S MANUAL.”

Relevant systems: T5593, T5592, T5586, T5585, T5376, T5375, and T5371

2.5.4 Online Manual

The Online Manual has been added.

As a result, the Help Window was deleted.

For more information, refer to “ATLworks TOOL URER’S MANUAL.”

Relevant systems: T5592, T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5376, T5375, T5371, T5336, and T5311

2.6 HTML Online Manual

The Online Manual user interface has been changed based on the Netscape Navigator(6.2.3). Accordingly, the main Online Manuals are now in the HTML version. Both the HTML and PDF versions can be used.

The Netscape Navigator(6.2.3)-based Online Manuals run in Solaris 7 and Solaris 8 of the SUN OS or in a Windows PC.

The Online Manuals provided in the HTML version are as follows:

- ATL-51 TEST PLAN PROGRAM REFERENCE MANUAL
- ATL-51 SOCKET PROGRAM REFERENCE MANUAL
- ATL-51 SCRAMBLE PROGRAM REFERENCE MANUAL
- ATL-51 PATTERN PROGRAM REFERENCE MANUAL (T5500)
- ATL-51 PATTERN PROGRAM REFERENCE MANUAL (T5376/T5375/T5371/T5336/T5311)
- ASX/U-51 HYBRID CONTROLLER USER'S MANUAL
- ASX/U-51 HYBRID CONTROLLER REFERENCE MANUAL
- T5500 SERIES TESTER UTILITY PROGRAM REFERENCE MANUAL
- T5500 SERIES CONTROLLER COMMAND REFERENCE MANUAL
- ERROR CODE TABLE REFERENCE MANUAL
- ATLworks TOOL USER'S MANUAL
- ASX/U-51 Communication Control Library Manual
- ADVANsite System Introduction
- ADVANsite TEST SYSTEM SOFTWARE INSTALLATION/SYSTEM START USER'S MANUAL VOL1 BASIC OS INSTALLATION
- ADVANsite TEST SYSTEM SOFTWARE INSTALLATION/SYSTEM START USER'S MANUAL VOL2 TESTER ENVIRONMENT SOFTWARE INSTALLATION
- ASX/U-51 BASIC SYSTEM SOFTWARE SOFTWARE SPECIFICATIONS
- Online Manual Reference Manual (HTML)

The Online Manuals provided in the PDF version are as follows:

- ACLI COMMANDS REFERENCE MANUAL (UNIX VERSION)
- T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION
- ASX/U-51 SYSTEST TESTUTL ATLworks UPDATE NOTICE
- Online Manual Reference Manual

For more information, refer to "Online Manual Reference Manual (HTML)."

Relevant systems: T5593, T5592, T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5376, T5375, T5371, T5336, and T5311

3. IMPROVED ITEMS

3.1 Tester Driver Controller

3.1.1 AFM REMOVE Statement

When the DUT numbers, which are assigned to SINGLE in the AFM MULTIMODE statement, and the DUT numbers, which are assigned to child E to H in the AFM DUTSEL statement, were specified, and then the fail addresses were stored in the AFM, if the AFM REMVOE statement was executed, The OR data of the fail store mode and mask mode was written not only in the AFM, where the mask mode was specified but also in the AFM, where the fail store mode was specified.

Improved ➡ The OR data of the fail store mode and mask mode is now written in the AFM, where the mask mode is specified. The data is no longer written in the AFM, where the fail store mode is specified.

Relevant revisions: REV 2.00 through REV 6.00C1

Relevant systems: T5581 and T5581H

3.1.2 SELECT OFFSET Statement and SELECT NOFFSET Statement

If a socket program, in which 31 or smaller was specified to the DR pin MAX or CP pin MAX, was used, the controller stopped at the SELECT OFFSET ALL statement or at the SELECT NOFF-SET ALL statement. If CTRL/C was entered at this point, a panic error (code=117) occurred.

Improved ➡ The controller now runs correctly.

Relevant revisions: REV 5.02 through REV 6.01

Relevant systems: T5593, T5592, T5586, T5585, T5376, and T5375

3.1.3 MEAS DC Statement and START DC Statement

(1) If a pin, which was not written in the DUT# line, was specified, or a pin, which did not exist in the system, was specified in a socket program, the program targeted these unspecified pins as test pins.

Improved ➡ A runtime error (code=3443) now occurs in the above stated situation.

Relevant revision: REV 6.01

Relevant system: T5593

(2) The error code (code=3443), which was used if the pin number exceeded the DR pin MAX or CP pin MAX in a socket program, was incorrect.

Improved ➡ A runtime error (code=12363) now occurs in the above stated situation.

Relevant revision: REV 6.01

Relevant system: T5593

3.1.4 SEND DBM, SEND CBM, and SEND DBMR Statements

If a pattern program, which generated a runtime error when it was transferred, was modified and compiled, and then transferred again in the controller, a runtime error occurred.

Improved ➡ The pattern program is now transferred correctly.

Relevant revision: REV 6.01

Relevant system: T5593

3.1.5 RESET DBM FILE Statement

The error code (code=12537), which was used if a file name, which was not transferred, was specified in the RESET DBM FILE statement, was incorrect.

Improved ➡ A runtime error (code=12572) now occurs in the above stated situation.

Relevant revision: REV 6.01

Relevant system: T5593

3.1.6 SET REJECTION Statement, SET/RESET EXCLUSION Statements, and SET/RESET EXCLUSION2 Statements

If the program was branched to a label whose name had been specified in one of the SET REJECTION, SET/RESET EXCLUSION, and SET/RESET EXCLUSION2 statements that had been executed in the routine called by the IF PASS/FAIL syntax or by ON FCTF, ON DCTF, ONCALL FCTF or ONCALL DCTF, no operation was performed for target DUTs in the following statements.

- SET REJECTION statement
- SET SORT statement
- SET/RESET EXCLUSION statement
- SET/RESET EXCLUSION2 statement
- SET/RESET CATEGORY statement
- SET/RESET CNT statement
- SET/RESET WFCNT statement
- SET/RESET FAIL statement
- SET/RESET RESULT statement

Improved ➡ The process is now performed correctly for the target DUTs.

Relevant revision: REV 5.05 through REV 6.01

Relevant system: T5593, T5376, and T5375

3.1.7 SEND SCRDATA Statement

If two scramble programs, which the ARIRAM data setting statement is specified for, were transferred in the order described below, the ARIRAM data for the second scramble program was not transferred.

The first program transfer		The second program transfer
SEND SCRDATA file-A	→	SEND SCRDATA file-B
SEND SCRDATA MAIN file-A	→	SEND SCRDATA file-B
SEND SCRDATA SUB file-A	→	SEND SCRDATA file-B

Improved ➡ The ARIRAM data for the second scramble program is now transferred.

Relevant revision: REV 6.01

Relevant system: T5593

3.1.8 WAIT TIME Statement

If the WAIT TIME statement was executed after executing the START MPAT *, a system error (code = 200) occurred although the system operated correctly.

Improved ➡ No system error occurs.

Relevant revision: REV 5.00 through REV 6.01

Relevant system: T5593, T5592, T5586, T5585, T5376, T5375, and T5371

3.2 Hybrid Controller

3.2.1 C++ new Operator

If the memory space, which was allocated by using the new operator, was not freed explicitly in a C++ program when the setting to start (boot) the tester processor from the network was specified, the memory leak occurred and the system stopped responding.

Improved ➡ By applying the solution described below, the memory space, which is allocated by using the new operator, is now freed automatically when the program is complete. As a result, no memory leak occurs and the system responds correctly.

[Solution]

Use the procedure below to change the setting so that the tester processor starts (boots) from the embedded flash memory.

(1) Open the tester processor console.

- When the tester processor serial cable is connected to serial port A:

```
% tip -9600 /dev/term/a
```

- When the tester processor serial cable is connected to serial port B:

```
% tip -9600 /dev/term/b
```

(2) Execute the flash boot setting script.

```
→ < boot-flash
```

(3) Restart the tester processor.

```
→ [CTRL]-[X]
```

(4) Close the console to complete the setting procedure.

```
→ ~~.
```

Relevant revisions: REV 6.00C through REV 6.01

Relevant systems: T5593, T5592, T5586, T5585, T5376, T5375, and T5371

3.2.2 AT_read_cnt

Even though the test fail counter space was allocated correctly, 1 was returned as the return value.

Improved ➡ 0 is now returned.

Relevant revisions: REV 6.00-1 through REV 6.01

Relevant systems: T5593, T5592, T5586, T5585, T5376, T5375, T5371

3.2.3 AT_set_rejection(), AT_set_rejectionm(), AT_set_exclusion(), AT_set_exclusionm(), AT_set_exclusion2(), AT_set_exclusionm2(), AT_reset_exclusion2(), AT_reset_exclusionm2()

If one of the following tester access libraries was executed after any of the SET REJECTION, SET EXCLUSION, or SET EXCLUSION2 statement was executed in the routine called by the IF PASS/FAIL syntax or by ON FCTF, ON DCTF, ONCALL FCTF or ONCALL DCTF, the specified DUTs were not excluded from the test target in the AT_set_xxxx library, and the specified DUTs were not included in the test target in the AT_reset_xxxx library.

AT_set_rejection(), AT_set_rejectionm(), AT_set_exclusion(), AT_set_exclusionm(), AT_set_exclusion2(), AT_set_exclusionm2(), AT_reset_exclusion2(), AT_reset_exclusionm2()

Improved ➡ The specified DUTs are excluded from and returned to the test target correctly.

Relevant revisions: REV 6.00 through REV 6.01

Relevant systems: T5593, T5376, and T5375

3.3 Tester Utility Program

3.3.1 Pattern Tracer (/MTRACE)

After the pattern tracer was executed, the ARIRAM setting was sometimes ignored.

Improved ➡ The ARIRAM setting is now enabled.

Relevant revisions: REV 1.00 through REV 6.00C1

Relevant systems: T5592, T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5376, T5375, T5371, T5336, and T5311

3.3.2 Data Scan (/DS)

If the offset data was specified to the ACLK, BCLK, CCLK, DREL, DRET, STRB, WSTRB, or TG-CLK and then the data scan was executed to the unit, the incorrect data was displayed.

Improved ➡ The correct data is now displayed.

Relevant revisions: REV 5.02 through REV 6.00C1

Relevant systems: T5592, T5376, and T5375

3.3.3 Parallel DFM Utility (/PDFMDISP)

When the Parallel DFM utility was not set, if /PDFMDISP was executed, an error message was incorrectly displayed on the top of the screen.

Improved ➡ The error message is now correctly displayed.

Relevant revisions: REV 6.01

Relevant systems: T5593

3.3.4 Program which Controls the Handler by Using the GPIB Interface (/STHAND)

The handler sometimes did not start.

Improved ➡ The handler starts correctly and performs the measurement.

Relevant revisions: REV 5.05 through REV 6.00C1

Relevant systems: T5376 and T5375

3.3.5 MARGIN Program (/MARGIN)

(1) If the margin test for the VT, which is connected to the driver channels, was executed, the margin was measured by using the comparator VT value set in the test plan program.

Improved ➡ If the margin test for the VT, which is connected to the driver channels, is executed, the margin is measured by using the driver VT value set in the test plan program.

Relevant revision: REV 6.01

Relevant system: T5593

(2) If the margin test for the VT, which is connected to the I/O channels, was executed, the VT margin was measured by using both the driver and comparator VT values.

Improved ➡ If the margin test for the VT, which is connected to the I/O channels, is executed, the VT margin is measured by using only the comparator VT value.

Relevant revision: REV 6.01

Relevant system: T5593

(3) If the program was returned to the test plan program after the VT margin test, the comparator VT value set in the test plan program was used as the driver VT set value.

Improved ➡ After the VT margin test, the driver VT value is reset to the value set in the test plan program.

Relevant revisions: REV 4.00 through REV 6.01

Relevant systems: T5593, T5592, T5591, and T5591R

3.3.6 Pin Monitor (/PM) and Extension Pin Monitor (/EPM)

If the following conditions (a) through (c) were all satisfied and DVTn was specified in the pin statement for high-speed clock channels (55, 56, 119, 120, 183, 184, 247, and 248), OUTL and VT were displayed in the Pin monitor or Extension Pin monitor.

Condition:

(a) When defining DUT pin names in a socket program, DUT pin numbers are defined by using PD for high-speed clock driver channels (55, 56, 119, 120, 183, 184, 247, and 248).

Example:

```
SOCKET  SOC1

  PD1=55
```

(b) DUTn is specified in the pin statement for DUT pin numbers that have been specified for high-speed clock driver channels (55, 56, 119, 120, 183, 184, 247, and 248) in the socket program.

Example:

```
PRO PRO1 SOC1

  PD1=DVT1,IN1,BCLK1,CCLK1,RZX,HSCMAIN
```

(c) The unit name is not specified in the Pin monitor, the page name is not specified in the Extension Pin monitor when Pn is specified as the unit name, or when PIN is specified as the page name.

Improved ➡ Both monitors now displays data as specified in the pin statement.

Relevant version: REV 6.01

Relevant systems: T5593

3.3.7 Fail Memory Data Transfer (/FMXFER)

If the target DBM test pattern program, which was transferred by using the SEND DBMR statement, was cleared by specifying the file name in the RESET DBM statement, a runtime error (code = 12537) occurred if FMXFER was executed.

Improved ➡ No runtime error occurs and FMXFER is executed.

Relevant version: REV 6.01

Relevant systems: T5593

3.4 ATLworks

3.4.1 CPANEL and ASXDebugger

If the controller operation mode was set to the hybrid mode and the CTRL/C invalid setting program (/CKBRCNT) was set to DISABLE, then the Reset button was clicked, the program stopped.

Improved ➡ Clicking the Reset button is now ignored when the CTRL/C invalid setting program (/CKBRCNT) is set to DISABLE.

Relevant versions: REV 6.00 through REV 6.01

Relevant systems: T5593, T5592, T5586, T5585, T5376, T5375, and T5371

3.4.2 ASXDebugger

In the hybrid debug mode, if a runtime error occurred in an ATL program and the following operations were performed in order, the program terminated.

[Operation]

(1) Reset

(2) Start

(3) After the program stops at test_main(), a breakpoint is set between test_main() and ATL_init(), and then Continue.

(4) After the program stops at the breakpoint, Continue.

Improved ➡ The program now runs correctly.

Relevant versions: REV 6.00A through REV 6.01

Relevant systems: T5593, T5592, T5586, T5585, T5376, T5375, and T5371

4. DELETED FUNCTIONS

4.1 ICD/U-51 and MPEDIT/U-51

ICD/U-51 and MPEDIT/U-51 no longer run in REV 6.02 and later.